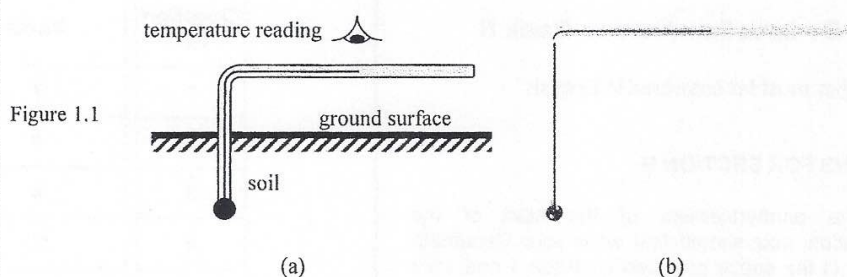


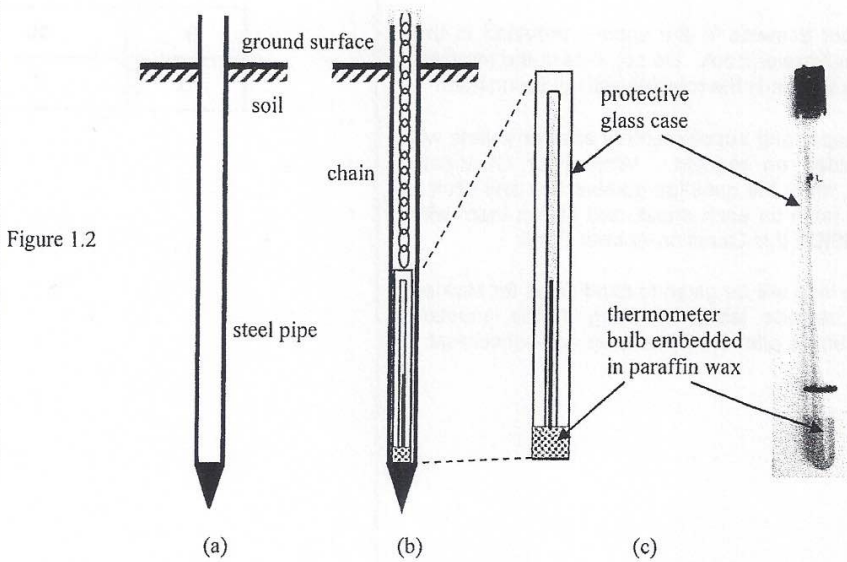
Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. Read the following passage about **soil thermometer** and answer the questions that follow.

The temperature of soil changes with depth, and this information is important to farmers and scientists. To measure soil temperatures at depths close to the ground surface, the bulb of a thermometer is buried in the soil. The stem of the thermometer is bent 90° for easy reading. Figure 1.1a is a schematic diagram and Figure 1.1b shows a photo of a soil thermometer.



For depths greater than 30 cm, a steel pipe is driven into the soil (Figure 1.2a); and a liquid-in-glass thermometer with a protective glass case is lowered into the steel pipe (Figure 1.2b). The bulb of the thermometer is embedded in paraffin wax (Figure 1.2c). To read the temperature, the thermometer is lifted out of the steel pipe by pulling the chain.



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- (a) As shown in Figure 1.1b, the bulb of the soil thermometer is very large compared to those of common thermometers. Suggest a reason for this design. (1 mark)

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- (b) On a certain morning, the air temperature is 15°C . An observer takes a measurement of the soil temperature at 1 m deep. The thermometer reading is 20°C . It is given that the mass of the paraffin wax enclosing the thermometer bulb is 0.015 kg, and the specific heat capacity of paraffin wax is $2.9 \times 10^3 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$.

- (i) Calculate the energy loss of the paraffin wax as it cools down to the air temperature. (2 marks)

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- (ii) It is known that the paraffin wax enclosing the bulb of the thermometer gains or loses energy at a constant rate of 0.5 J s^{-1} , estimate the time taken for the paraffin wax to reach the air temperature after the thermometer is lifted out of the soil. (2 marks)

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- (iii) If there is no paraffin wax enclosing the bulb of the thermometer, explain how the thermometer reading as recorded by the observer is affected. (2 marks)

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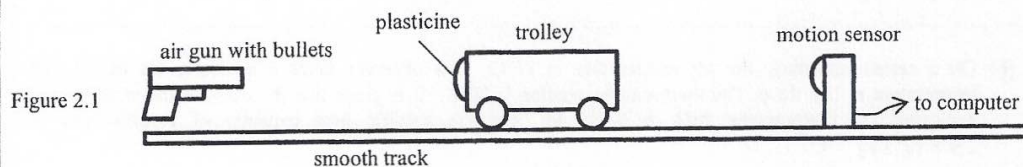
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2. The following experimental items are provided to set up an experiment to estimate the speed of a bullet fired from an air gun.

a smooth track
 a trolley
 a motion sensor used to measure the speed of the trolley
 some plasticine
 an air gun and bullets
 an electronic balance

The set-up is shown in Figure 2.1.



Describe the procedures of the experiment. State the physical quantities to be measured and an equation for finding the speed of the bullet. Write down **ONE** precaution for getting a more accurate result. (5 marks)

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- *3. The average kinetic energy of one monatomic gas molecule at temperature T is given by

$$E_K = \frac{3}{2} \left(\frac{R}{N_A} \right) T,$$

where R is the universal gas constant and N_A is the Avogadro constant. A monatomic gas is heated from 300 K to 350 K under fixed volume.

- (a) Estimate the ratio of the root-mean-square speed ($c_{r.m.s.}$) of the gas molecules at the two temperatures
 $\left(\frac{c_{r.m.s.} \text{ at } 350 \text{ K}}{c_{r.m.s.} \text{ at } 300 \text{ K}} \right).$ (2 marks)

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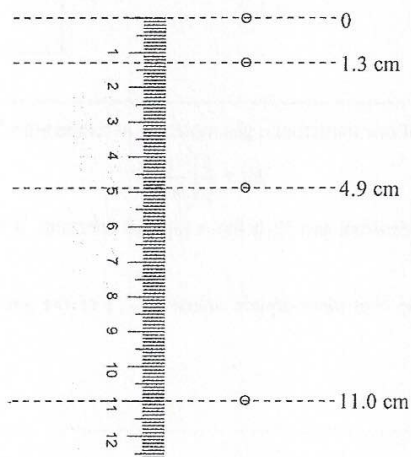
- (b) Thus, using kinetic theory, explain why the gas pressure would increase. (2 marks)

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4. (a) A steel ball bearing is released from rest at time $t = 0$. A stroboscopic photo is taken at 0.05 s time intervals. The results are shown in Figure 4.1. Neglect air resistance.

Figure 4.1



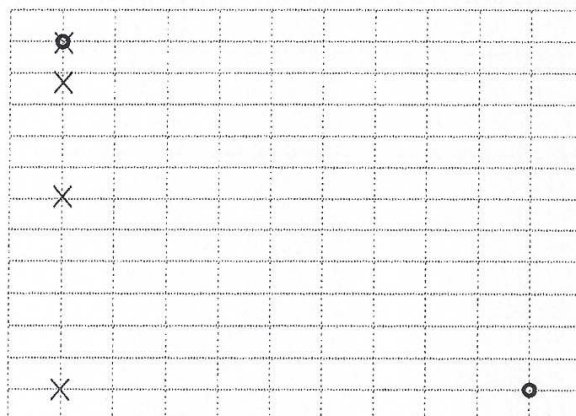
- (i) Estimate the acceleration due to gravity using the data in Figure 4.1.

(2 marks)

Answers written in the margins will not be marked.

- *(ii) The bearing is now projected horizontally instead of released from rest. The bearing is projected at time $t = 0$, and a stroboscopic photo is taken at 0.05 s time intervals. The first and the last image of the stroboscopic photo are shown using circles (●) in Figure 4.2. For reference, the stroboscopic photo of the bearing released from rest is also shown in the figure using crosses (x).

Figure 4.2



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- (1) In Figure 4.2, mark the positions of the projected bearing in the stroboscopic photo using circles (●). (2 marks)
- (2) Given that the bearing is projected horizontally with an initial speed of 1 m s^{-1} , use the results of (a)(i) to calculate the speed of the projected bearing when the last image was taken. (3 marks)

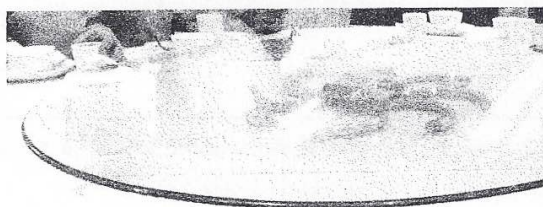
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- (b) If a small ball is released from rest from the top of a cliff, the speed of the ball becomes constant after a period of time. By considering the forces acting on the ball and using Newton's laws of motion, explain why the speed of the ball becomes constant. (3 marks)

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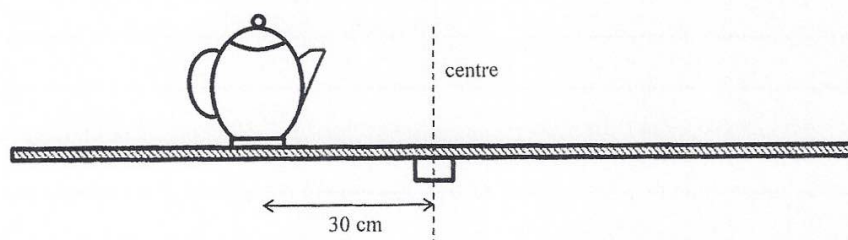
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*5. The photo shows a turntable commonly used in restaurants.



A teapot of mass 1 kg is put 30 cm from the centre of a horizontal turntable, Figure 5.1 shows the side view. When the turntable is rotating, the teapot remains at the same position on the turntable.

Figure 5.1



- (a) On Figure 5.1, draw and label all the forces acting on the teapot when the turntable is rotating. (2 marks)
- (b) Taking the teapot as a point mass, estimate the net force acting on the teapot when the turntable is rotating at a rate of 0.5 revolutions per second. (3 marks)

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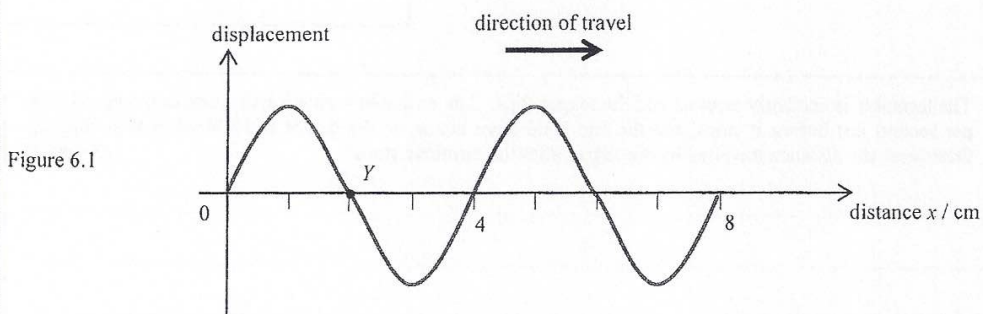
- (c) The turntable is suddenly stopped and the teapot slips. The turntable is rotating at a rate of 0.5 revolutions per second just before it stops, and the frictional force acting on the teapot is 10 N when it is slipping. Determine the distance travelled by the teapot after the turntable stops. (3 marks)

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6. (a) A dipper vibrating with a frequency of 5 Hz is put in a water tank. Figure 6.1 shows the displacement-distance graph of the water wave at time $t = 0$. Y is a particle in the water tank.

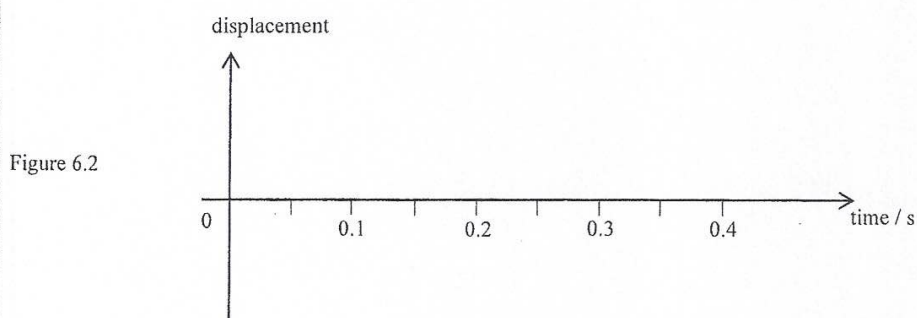


- (i) Determine the wave speed of the water wave. (2 marks)

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- (ii) State the direction of motion of particle Y at $t = 0$. (1 mark)

- (iii) Sketch the displacement-time graph of particle Y between $t = 0$ and $t = 0.4$ s in Figure 6.2. (2 marks)

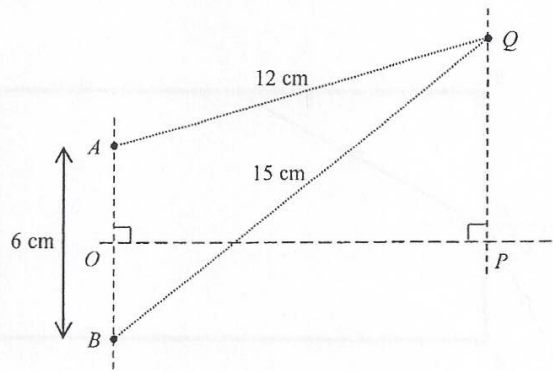


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- (b) In Figure 6.3, A and B are two dippers vibrating in phase in a water tank. The distance between A and B is 6 cm. OP is the perpendicular bisector of AB . Q is a second minimum from P , where $AQ = 12$ cm and $BQ = 15$ cm.

Figure 6.3



- (i) Explain why a minimum occurs at Q . (2 marks)

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- (ii) Determine the wavelength of the water wave. (2 marks)

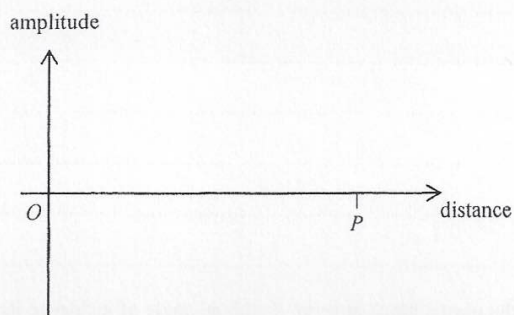
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- (iii) Sketch in Figure 6.4 how the **AMPLITUDE** of the water wave varies along the line OP . (1 mark)

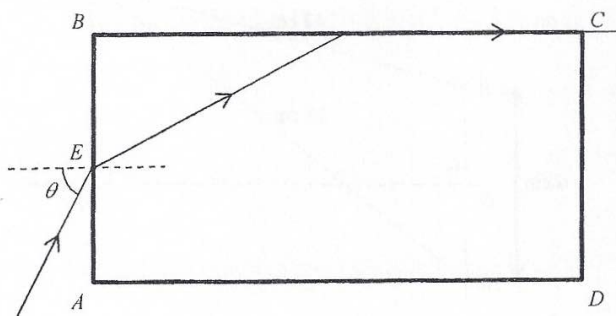
Figure 6.4



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7. (a) A light ray enters a rectangular plastic block $ABCD$ from air at point E , and the angle of incidence is θ . The light ray emerges along face BC as shown in Figure 7.1. The refractive index of the plastic is 1.36.

Figure 7.1



- (i) Find the critical angle of the plastic. (2 marks)

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- (ii) Find the value of θ . (3 marks)

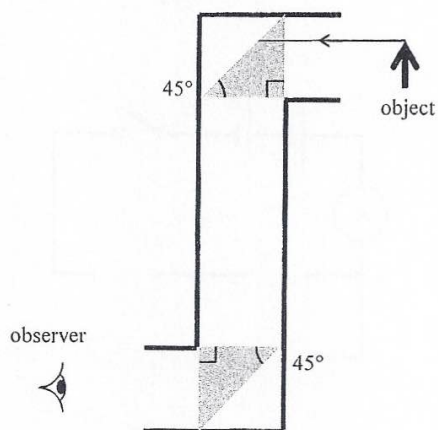
- (iii) If the light ray enters the plastic block at point E with an angle of incidence larger than θ , sketch the path of the light ray in Figure 7.1. (2 marks)

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Answers written in the margins will not be marked.

- (b) A student designs a periscope using two plastic prisms, the refractive index of the plastic is 1.36. As shown in Figure 7.2, an object is placed in front of the periscope.

Figure 7.2



- (i) Complete the path of the light ray from the object in Figure 7.2, and explain why the periscope fails to work. (3 marks)

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- (ii) What can be used to replace the two plastic prisms so that the periscope can work properly? (1 mark)

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8. A student uses the following apparatus to measure the resistance of a tungsten filament light bulb.

a battery, a switch, a variable resistor, an ammeter, a voltmeter, a light bulb

(a) Figure 8.1 shows an incomplete circuit for the experiment. The '+' symbol represents the positive terminal of the ammeter.

Use suitable circuit symbols to complete the circuit, and mark the positive terminal of the voltmeter with '+'. (3 marks)

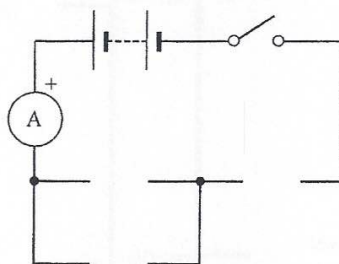


Figure 8.1

The table below and Figure 8.2 show the results obtained.

Voltage across the light bulb V / V	0	0.1	0.2	0.3	0.4	0.5	1.0	2.0	3.0
Current I / mA	0	76	112	126	133	139	170	226	273

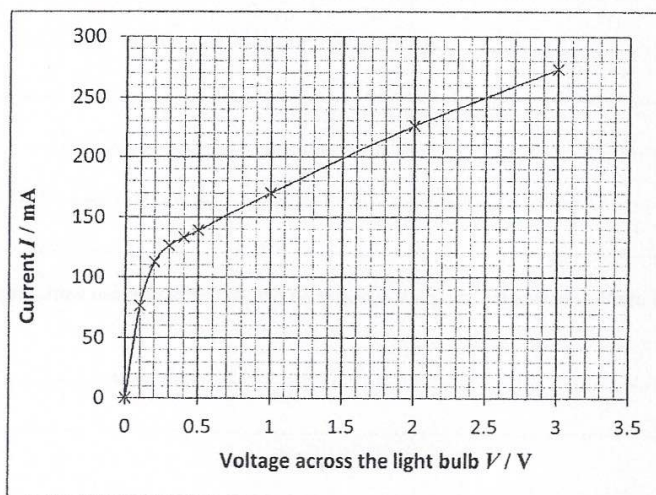


Figure 8.2

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- (b) Briefly explain the variation of the resistance of the light bulb with the voltage across the light bulb. (2 marks)

- (c) The student claims that since the resistance of the light bulb is not a constant, the equation $R = V/I$ cannot be used to calculate the resistance of the light bulb. Briefly explain why his claim is wrong. (1 mark)

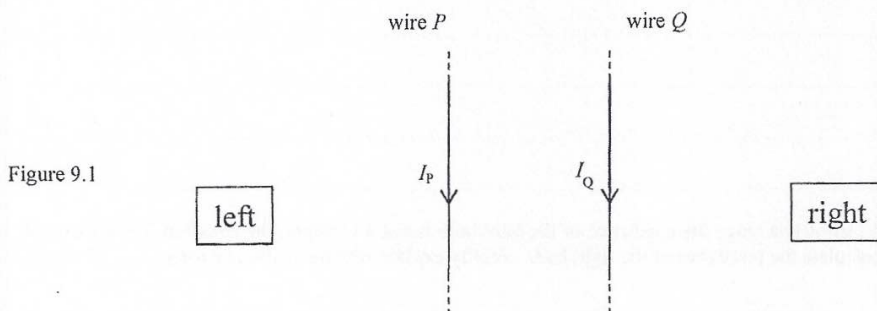
- (d) Determine the resistance of the light bulb at $V = 0.1 \text{ V}$ and 2.5 V . (3 marks)

- (e) It is given that the cross-sectional area of the tungsten filament in the light bulb is $1.66 \times 10^{-9} \text{ m}^2$, and the resistivity of tungsten at room temperature is about $5.6 \times 10^{-8} \Omega \text{ m}$. Estimate the length of the tungsten filament in the light bulb using the appropriate resistance found in (d). (3 marks)

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Answers written in the margins will not be marked.

9. (a) Two long straight current carrying wires, P and Q , are parallel to each other and lie on the plane of the paper as shown in Figure 9.1. The currents in the wires, I_P and I_Q , flow in the same direction.



- (i) State the direction (to the left / to the right / into the paper / out of the paper) of the magnetic field at Q due to P . (1 mark)

- (ii) In Figure 9.1, draw the direction of the magnetic force acting on Q due to P . (1 mark)

- (iii) Show that the magnitude of the magnetic force per unit length F_l acting on Q due to P is

$$F_l = \frac{\mu_0 I_P I_Q}{2\pi r},$$

where μ_0 is the permeability of free space and r is the separation between the two wires. (3 marks)

- (iv) For the magnetic force acting on Q due to P and the magnetic force acting on P due to Q , if $I_P \neq I_Q$, briefly explain whether the two forces are equal in magnitude. (2 marks)

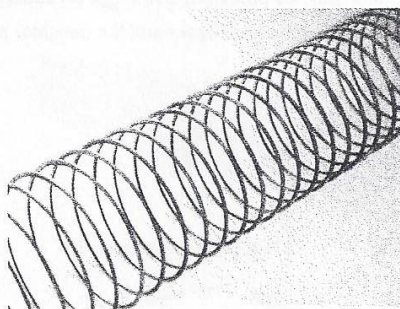
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(b) Figure 9.2 shows a metal slinky spring.

Figure 9.2



- (i) If a direct current passes through the spring, briefly explain whether the spring will be compressed or stretched due to magnetic force. (2 marks)

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- (ii) A student suggests that the spring will be compressed and stretched alternately due to magnetic force when an alternating current passes through. Briefly explain why he is wrong. (1 mark)

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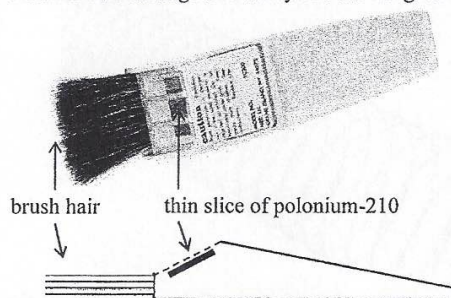
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10. Dust may adhere to the surfaces of photos and films due to electrostatic attraction. To remove the dust effectively, a special brush with a thin slice of polonium-210 ($^{210}_{84}\text{Po}$) fixed near the brush hair as shown in Figure 10.1 may be used. Polonium-210 undergoes α decay and the daughter nucleus lead (Pb) is stable.

Figure 10.1



- (a) Write a nuclear equation for the decay of polonium-210. (2 marks)

- (b) Briefly explain how the α particles help clean the charged dust. (2 marks)

- (c) Briefly explain why the polonium-210 slice must be fixed near to the brush hair. (1 mark)

- *(d) The manufacturer recommends that the brush should be returned to the factory for replacement of the polonium-210 slice every year. Taking the activity of a newly replaced polonium-210 slice as 1 unit, find its activity after one year (365 days). Given: half-life of polonium-210 is 138 days. (2 marks)

END OF PAPER

Sources of materials used in this paper will be acknowledged in the booklet *HKDSE Question Papers* published by the Hong Kong Examinations and Assessment Authority at a later stage.

Answers written in the margins will not be marked.